From Schools to Worlds - How Complementary AI Foundations Programs Can Close a World—and How a World Can Revise Itself

- How Complementary AI Foundations Programs Can Close a World—and How a World Can Revise Itself

A Research-Federation Map Through Natural Abstraction, Active Inference, Compositional World Modeling, Representation Geometry, and Embedded Agency

---

Abstract

Foundational research on artificial intelligence appears fragmented. One programme asks why useful abstractions arise from high-dimensional reality. Another studies perception and action under generative models. Another develops compositional formalisms for constructing coherent models. Another measures the geometry of internal representations. Another asks how an agent can reason when it is bounded, self-referential, and embedded inside the world it models.

These programmes are often discussed separately, and sometimes appear to compete for the status of a general theory of intelligence. This article proposes a different interpretation.

Their strongest contributions may be **complementary rather than substitutive**.

Natural Abstraction can contribute the effective variables of a world. Active Inference can give those variables operational dynamics. Compositional and categorical approaches can constrain how local models form a coherent formal whole. Representation Geometry and mechanistic measurement can test whether that formal structure is actually realized in an intelligent system. Agent Foundations can then place the observer inside the resulting structure, where boundedness forces the problem back toward abstraction.

This produces a closed research-complementarity cycle:

Abstraction → Dynamics → Composition → Realization → Embeddedness → Abstraction.

The central proposal is that this is more than a convenient map of research areas. Under suitable interfaces, the successive contributions can be interpreted as progressively completing an **Effective WORLD**:

$\mathcal{W} = (V,F,C,M,O).$ (1.1)

Here V denotes effective distinctions, F operational dynamics, C compositional structure, M measurable realization, and O the embedded observer relation.

The five research traditions do not map exclusively onto these five coordinates, nor does their ordering imply chronological dependence. Rather, each programme has developed unusually powerful machinery around one or more structural requirements of worldhood, while the unresolved boundary of one programme naturally exposes questions emphasized by another.

The article then connects this horizontal problem of **WORLD constitution** to the Five-Regime Boundary Circulation framework developed in *From Possibility to Revision*. An Effective WORLD is not yet a persistent or self-revising world. Historical trace, unresolved residual, dual ledgering, latching, and revision must also be added:

$\mathcal{W}^{R} = (\mathcal{W}, L^{+}, L^{-}, U; P)$. (1.2)

This yields two complementary forms of closure:

horizontal closure — enough structure exists for a bounded observer to inhabit an operational world;

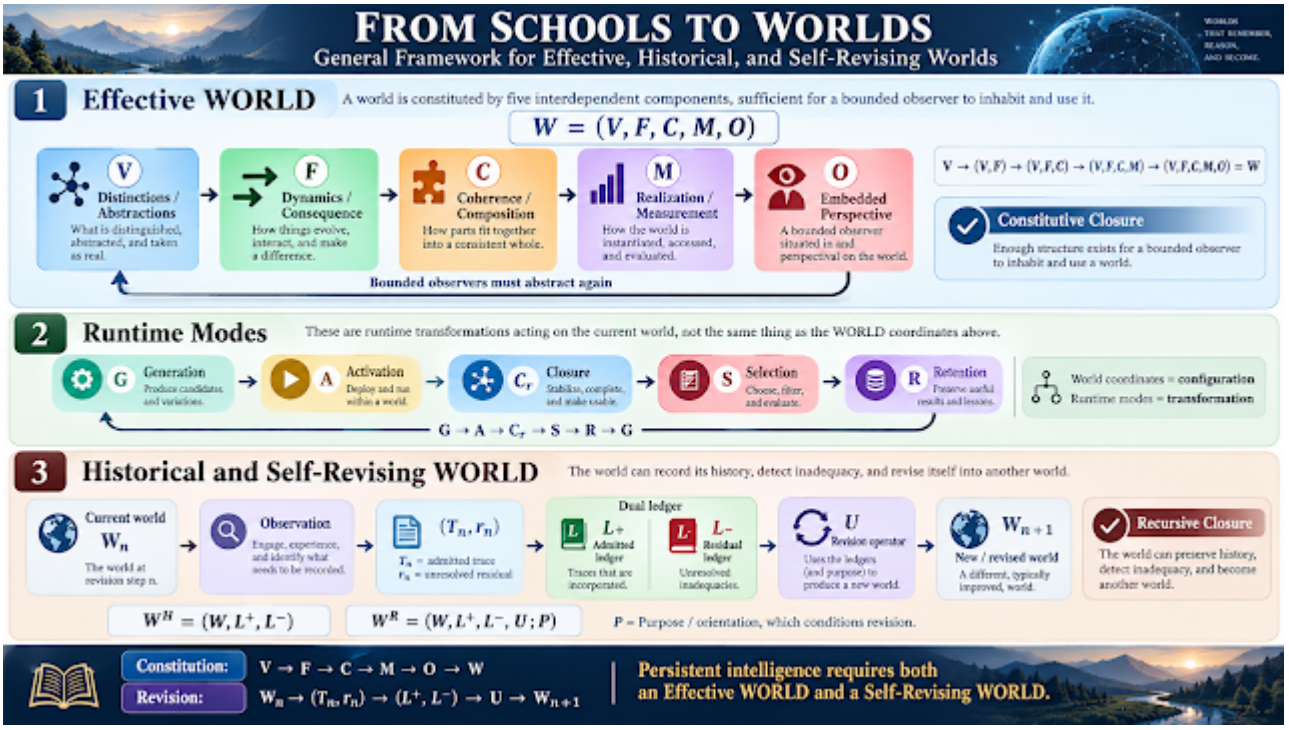
and

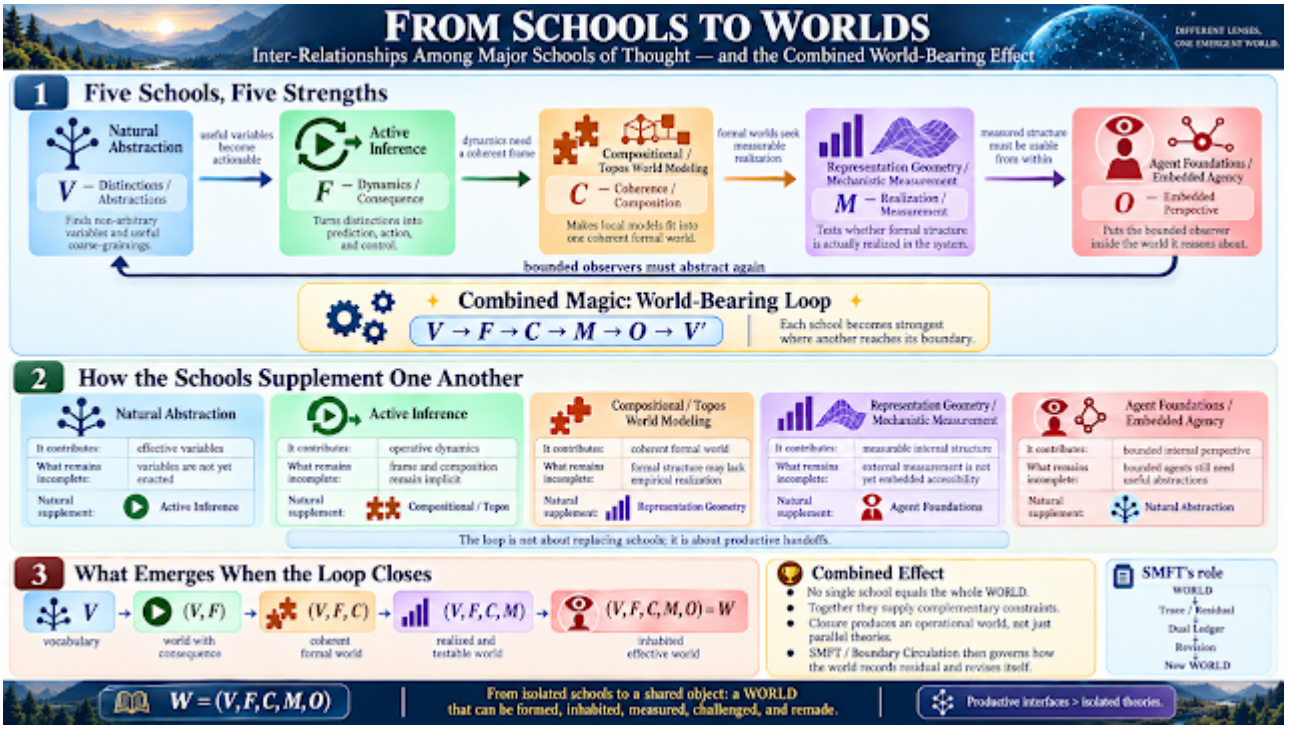
vertical closure — that world can accumulate history, recognize its own inadequacy, and become another world.

The resulting picture is not a claim that one framework subsumes the others. It is a proposal for a **research federation**: a common interface grammar in which different theoretical programmes can preserve their own mathematics while becoming mutually informative.

The central thesis is simple:

> **Each programme becomes especially valuable where another programme reaches its natural boundary: abstractions need dynamics, dynamics need coherent composition, formal worlds need measurable realization, realized structures need an embedded observer, and bounded observers need abstraction. Close those interfaces, and a collection of research programmes begins to look like a theory of how a WORLD is formed. Add trace, residual, and revision, and that WORLD becomes capable of becoming another world.**





---

1. The Fragmentation of AI Foundations May Be Partly an Interface Problem

Foundational AI research contains many deep programmes that appear, at first sight, to ask very different questions.

Natural Abstraction asks why a complex physical world may admit relatively low-dimensional summaries that different cognitive systems can converge upon. In its strongest form, the Natural Abstraction Hypothesis proposes that many cognitive systems may discover approximately the same high-level abstractions because the world itself supports such compressed summaries. Alignment Forum

Active Inference starts from a different direction. It develops a generative-model framework in which perception, belief updating, action, planning, and policy selection can be treated within a common inferential architecture. Current Active Inference materials explicitly characterize the framework in terms of generative models, perception as inference, learning as model update, action, and expected free energy. Active Inference Institute

Compositional world-modeling programmes ask another question again: how can complex models be constructed from parts while preserving formal interpretability and the ability to reason about interfaces? Topos Institute has explicitly pursued compositional world-modeling using category-theoretic methods, and more recently has emphasized the need for constrained but fluid formal frames that can be shifted when circumstances demand. Topos Institute

Representation Geometry and neighboring mechanistic approaches ask whether internal representations possess measurable structure: subspaces, trajectories, basins, transitions, persistent directions, or other geometric signatures.

Agent Foundations and Embedded Agency approach the problem from yet another level. MIRI's public Agent Foundations work characterized the central difficulty as reasoning and decision-making for agents that are smaller than their environment, embedded within it, able to reason about themselves, and lacking a crisp agent/environment boundary. Machine Intelligence

Research Institute

These programmes need not be five rival answers to one question.

They may instead be answers to **different questions required to complete one larger object**.

---

2. The Wrong Question: Which School Is the Master Theory?

A conventional comparison often asks:

> Which framework is most fundamental?

That framing immediately creates competition.

Should intelligence fundamentally be understood through:

abstraction?

Bayesian inference?

category theory?

geometry?

embedded agency?

The question may be badly typed.

These programmes often operate on different kinds of objects and at different explanatory levels.

A theory of abstraction may answer:

> What variables should exist in a useful model?

A theory of inference may answer:

> How should beliefs and actions evolve once such variables exist?

A compositional framework may answer:

> Under what conditions can multiple local models belong to one coherent formal structure?

A representation-measurement programme may answer:

> Is that proposed structure actually instantiated in the system?

An embedded-agency programme may answer:

> What changes when the agent itself is a bounded part of the structure it reasons about?

These questions can conflict.

But they need not.

A different research strategy is therefore possible:

> **Instead of ranking the programmes vertically, connect them horizontally by their unresolved interfaces.**

This article calls that strategy **research federation**.

---

3. Research Federation Rather Than Theory Replacement

A federation does not require its members to share one ontology.

Each research programme may retain:

- its own primitives;
- its own mathematical language;
- its own empirical standards;
- its own local questions;
- its own criteria of explanatory success.

The federation adds something else:

explicit interfaces.

Suppose programme A constructs some object X_A but leaves open a structural question R_A.

Programme B provides machinery H_B that acts on that boundary:

$H_B(X_A,R_A) \rightarrow X_B.$ (3.1)

Call A→B a **productive research handoff** when:

1. the output of A creates a meaningful input for a problem emphasized by B;
2. B adds a type of structure that was not explicit in A's output;
3. the enriched object supports operations that were previously unavailable.

This is not reduction.

It is **structural supplementation**.

Programme B need not be the only possible supplement to A.

Indeed, overlap is expected.

The claim is only that B contains unusually mature concepts for one unresolved interface.

---

4. From Theory Coverage to Object Completion

This suggests a different way of comparing research programmes.

Do not ask:

> Which theory covers the most territory?

Ask:

What kind of structure does this programme add to the object under construction?

Begin with effective distinctions:

V. (4.1)

Add operative dynamics:

(V,F). (4.2)

Add composition and admissibility:

(V,F,C). (4.3)

Add measurable realization:

(V,F,C,M). (4.4)

Finally add an embedded observer relation:

(V,F,C,M,O). (4.5)

Define:

$\mathcal{W}$ = (V,F,C,M,O). (4.6)

The symbol $\mathcal{W}$ will denote an **Effective WORLD**.

The capitalization is deliberate.

It does not mean metaphysical reality.

It means a sufficiently closed operational structure within which a bounded observer can:

distinguish states,

encounter lawful change,

compose local relations,

obtain stable traces,

and locate itself as a participant.

The five coordinates therefore answer five different questions:

V — Distinction: What can count as a variable?

F — Consequence: What can happen?

C — Coherence: What belongs together?

M — Realization: What differences actually leave stable traces?

O — Perspective: From where inside the structure can anything be known or acted upon?

An Effective WORLD requires all five types of structure.

It does not require that five separate theories own them.

---

5. Why the WORLD Tuple Is Not a Five-School Trick

The construction:

$\mathcal{W}$ = (V,F,C,M,O) (5.1)

must not be interpreted as:

five coordinates exist because we selected five research programmes.

That would simply reproduce the problem avoided in the Five-Regime paper.

The logical direction is instead:

1. ask what minimally distinguishes an operational world from a disconnected collection of models;
2. identify several necessary structural conditions;
3. only afterward ask which existing research programmes have developed unusually strong machinery around those conditions.

The mapping is therefore many-to-many.

A research programme p may be represented by a coverage profile:

w(p) = (wV,wF,wC,wM,wO). (5.2)

There is no requirement that:

wV=1

for Natural Abstraction,

or:

wF=1

for Active Inference.

Indeed, such assignments would be false.

Natural Abstraction also concerns prediction and observer convergence.

Active Inference includes representation learning, generative structure, preference, policy, and model update.

Compositional methods can influence abstraction itself.

Representation measurement applies across the entire world tuple.

Embeddedness constrains every coordinate.

The point of the map is therefore not exclusivity.

It is **dominant research concentration**.

---

6. Research Programme I — Natural Abstraction: Giving the WORLD Distinctions

Consider a high-dimensional environment:

E. (6.1)

A bounded observer cannot retain every microscopic degree of freedom.

It needs some mapping:

$\mathcal{A} : E \to V$, (6.2)

where V has much lower effective complexity than E.

Natural Abstraction asks whether some such mappings are privileged by the structure of the world rather than chosen arbitrarily.

One formulation of the Natural Abstraction Hypothesis proposes that physical systems often have relatively low-dimensional summaries relevant at a distance, and that diverse cognitive architectures may therefore converge on approximately the same high-level abstractions. Alignment Forum

In the present framework, its strongest contribution can be summarized as:

$E \to V$. (6.3)

It gives the WORLD **distinctions**.

Examples might include:

object,

temperature,

agent,

resource,

boundary,

market,

predator,

tool,

promise,

cause.

The particular examples are secondary.

The deep problem is:

> Why should a bounded intelligence carve the environment at these joints rather than infinitely many others?

A WORLD cannot begin without distinctions.

---

7. Natural Abstraction Does Not Yet Give a WORLD

Suppose Natural Abstraction succeeded perfectly.

The system possesses a set of excellent variables:

$V = \{v_1,\ldots,v_k\}$. (7.1)

That is still not enough.

Variables do not tell us:

how states change,

which interventions matter,

how observations update beliefs,

what actions are available,

or which future states are preferred.

A taxonomy is not yet a world.

An abstraction becomes significantly more powerful when it participates in consequence.

Thus the unresolved boundary is:

$V \rightarrow ?$. (7.2)

The next question is:

> **What happens through these distinctions?**

This creates the first productive handoff.

---

8. Handoff I — From Distinction to Consequence

The transition is:

Natural Abstraction → Active Inference. (8.1)

Again, the arrow does not mean that Active Inference logically depends on a separate Natural Abstraction module.

Active Inference can contain its own latent-variable learning, hierarchical representation, and model adaptation.

The handoff identifies a narrower complementarity:

> Natural Abstraction sharpens the question of which variables are non-arbitrary;
> Active Inference supplies a mature architecture for turning represented variables into prediction, inference, action, and policy.

Thus:

$V \rightarrow (V,F).$ (8.2)

The object gains dynamics.

This is the first step from:

a vocabulary

toward:

an operative WORLD.

---

9. Research Programme II — Active Inference: Giving Distinctions Consequence

Active Inference describes agents in terms of generative models relating hidden states, observations, actions, beliefs, and policies.

Current Active Inference descriptions explicitly distinguish the **generative model**—the agent's probabilistic model of hidden causes—from the **generative process**, the external causal process producing observations. Perception updates beliefs under the generative model, while action changes the agent's relation to the generative process. Active Inference Institute

The relevant contribution here can be represented schematically as:

$F : V \times U \rightarrow V,$ (9.1)

or probabilistically:

$P(v_{t+1} \mid v_t,u_t).$ (9.2)

If interventions are represented explicitly:

$F = \{P(v'|v), P(v'|do(u),v), \ldots\}.$ (9.3)

Then F includes not merely passive change but **action-conditioned consequence**.

In the present positioning framework, Active Inference therefore contributes strongly to the second WORLD coordinate:

F.

It gives the distinctions of V a life.

They become things that can:

change,

predict,

surprise,

guide action,

and participate in policy.

10. Why Active Inference Is More Than “Activation”

It would be tempting to map:

Active Inference = Activation.

That would be too crude.

The Five-Regime architecture from the companion article distinguishes runtime modes:

$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G.$ (10.1)

Active Inference can contribute to several of them:

belief formation can participate in Generation;

action and policy strongly touch Activation;

model evidence and selection touch Selection;

learning and updating can alter retained model structure.

Therefore the research-programme map and the Five-Regime runtime must remain separate.

A useful distinction is:

> **The WORLD coordinates describe what structural ingredients an effective world contains. The Five-Regime modes describe what kinds of transformation dominate while the system operates and revises that world.**

Thus:

WORLD coordinates ≠ runtime regimes. (10.2)

This distinction will become increasingly important as the research programmes are connected.

---

11. Where Active Inference Reaches Its Next Interface

Suppose the agent has:

variables V

and operative dynamics F.

It can predict and act.

Yet a deeper problem remains.

The generative model contains:

states,

observations,

policies,

contexts,

hierarchies,

interfaces,

and possibly several local models.

What makes these components parts of **one coherent formal world**?

When can models compose?

When are contexts compatible?

What must remain invariant under a change of frame?

When should one formal frame be replaced by another?

These questions move beyond dynamics alone.

They concern:

composition.

The next unresolved transition is therefore:

$(V,F) \rightarrow (V,F,?)$. (11.1)

The missing coordinate is:

C.

And this creates the second major handoff:

> **Dynamics need a world in which to be dynamics.**

---

12. Handoff II — From Consequence to Coherence

This brings Active Inference into contact with compositional and category-theoretic world-modeling.

Topos Institute's compositional world-modeling programme explicitly seeks formal frameworks in which complex models can be constructed and analyzed compositionally, with category theory proposed as a useful mathematical language for organizing such models. Topos Institute

More recently, Topos has framed a related problem in terms of **formal frames**: modern LLM-assisted modeling lowers the cost of producing formal artifacts, but increases the need for constraints, reliability, and the ability to shift frames when circumstances demand. Topos Institute

This is almost exactly the interface we require.

Active Inference asks:

> How does an agent operate under a generative structure?

Compositional world-modeling asks:

> How do the pieces of such structures fit together, and how may the frame itself be changed without destroying formal intelligibility?

Thus:

(V,F) → (V,F,C). (12.1)

The WORLD gains **coherence**.

---

13. The First Three Steps

We can already see an emerging construction:

High-dimensional environment
→ abstraction
→ V

V
→ inference/action
→ (V,F)

(V,F)
→ composition/frame
→ (V,F,C). (13.1)

Or in words:

Distinguish → Enact → Compose.

At this point we have something increasingly world-like.

There are meaningful variables.

They have dynamics.

Their local relations can belong to a larger formal structure.

But one major danger remains.

A formally coherent world can exist entirely on paper.

The next question is therefore unavoidable:

> **Is this structure actually realized in the intelligent system?**

That question takes us from formal worldhood to empirical worldhood.

And it creates the third handoff.

14. Research Programme III — Representation Geometry: Giving the Formal WORLD a Measurable Realization

Suppose we have reached:

$\mathcal{W}$_formal = (V,F,C). (14.1)

The system now possesses:

effective distinctions,

operative dynamics,

and a coherent formal structure.

Yet this still leaves open a decisive question:

Where is this world actually realized?

A formal decomposition can be internally elegant without corresponding to the organization of the system that supposedly implements it.

If we claim that an AI has:

a Gate,

a stable attractor,

a latched declaration,

a residual accumulation process,

or a transition between effective worlds,

then those structures should eventually leave measurable signatures.

This motivates the fourth WORLD coordinate:

M = measurable realization. (14.2)

14.1 From Formal Structure to Empirical Structure

Let:

M : $\mathcal{W}$_formal → Y (14.3)

be a family of measurement maps into some empirically accessible space Y.

Depending on the system, Y might include:

- activation trajectories;
- representational subspaces;
- circuit states;
- persistent memory traces;
- behavioral interventions;
- state-transition statistics;
- geometric or dynamical observables.

The important point is not that every world coordinate must correspond to a single neuron, vector, or latent axis.

Rather:

> **differences that matter within the effective WORLD should generate stable differences somewhere in the realization.**

If two purported world states:

$v_1 \neq v_2 \quad (14.4)$

produce no distinguishable consequence in any relevant measurement or intervention, their status as separate effective variables becomes questionable.

---

15. Representation Geometry as an Experimental Interface

The earlier SMFT comparison already suggested a particularly concrete bridge between structural hypotheses and measurable geometry:

Gate → basin crossing?

Latching → hysteresis?

Residual accumulation → curvature or error growth?

Re-declaration → manifold or subspace switch?

Trace stabilization → persistent representation? 探討 SMFT 串聯其它名門正派成為五行流轉的可行性

These are not established equivalences.

They are **measurement hypotheses**.

That distinction is crucial.

The productive role of Representation Geometry is therefore not:

> prove the Five-Regime theory by finding a pentagon in activation space.

It is:

> **translate structural claims into observables that can support, constrain, or falsify them.**

This makes Representation Geometry—or more broadly, mechanistic measurement—an important complement to formal world construction.

---

16. Handoff III — From Coherence to Realization

The third productive handoff can now be written:

Compositional World Modeling → Representation Geometry. (16.1)

Or structurally:

$(V,F,C) \rightarrow (V,F,C,M). \quad (16.2)$

The supplementary progress is:

formal coherence
→ empirical realization.

In words:

> **A formal world becomes scientifically stronger when its distinctions and transitions can be located in the system that implements it.**

This gives the WORLD its fourth structural requirement.

We now have:

Distinction → Consequence → Coherence → Realization. (16.3)

But something is still missing.

A measured system is not yet necessarily an **inhabited** system.

---

17. External Measurement Is Not Yet an Embedded Perspective

Suppose a researcher discovers that a model contains:

- a stable low-dimensional subspace;
- a recurrent trajectory;
- an attractor basin;
- a sharp transition;
- a persistent representation.

These are valuable findings.

But the researcher occupies an external position.

The next question is fundamentally different:

> **Which of these structures are available to the agent itself?**

An external scientist may possess information the agent does not.

The scientist may identify:

a hidden state,

a circuit,

an attractor,

or a causal dependency

that the agent cannot represent explicitly.

Therefore:

measurable realization ≠ embedded accessibility. (17.1)

This distinction introduces the fifth WORLD coordinate.

---

18. Research Programme IV — Agent Foundations and Embedded Agency: Putting the Observer Inside the WORLD

The relevant Agent Foundations tradition emphasizes that realistic intelligent agents do not stand outside the world they reason about.

They are:

bounded,

smaller than their environment,

part of the environment,

and potentially required to reason about themselves.

The earlier comparison summarized this research concentration through issues such as:

logical uncertainty,

embedded agency,

self-reference,

decision theory,

counterfactual reasoning,

bounded rationality,

the agent/environment boundary,

and reflective agents. 探討 SMFT 串聯其它名門正派成為五行流轉的可行性

This changes the WORLD problem radically.

The observer cannot simply be written as:

O_external : WORLD → measurement. (18.1)

Instead:

$O \in \mathcal{W}$ (18.2)

in the functional sense that the observer's own computational and representational capacities are themselves part of the world being modeled.

---

19. The Fifth Coordinate: Embedded Perspective

Let:

$O : M \to Z_O$ (19.1)

represent the information actually available to the bounded observer.

Here Z_O need not equal the full measurable state M.

Usually:

$\text{Information}(O) < \text{Information}(M)$. (19.2)

The observer encounters only a constrained projection of the realization.

Thus O determines:

- what can be sensed;
- what can be represented;
- what can be remembered;
- what can be acted upon;
- what can count as evidence;
- what can enter self-reference.

This gives the Effective WORLD its fifth component:

$\mathcal{W} = (V,F,C,M,O)$. (19.3)

At this point the object is no longer merely formal or measurable.

It has an **internal standpoint**.

---

20. Handoff IV — From Realized Structure to Inhabited Structure

The fourth productive handoff is:

Representation Geometry → Agent Foundations. (20.1)

Structurally:

$(V,F,C,M) \to (V,F,C,M,O)$. (20.2)

The supplementary progress is:

measured structure
→ embedded perspective.

Or more succinctly:

> **Representation Geometry asks where the structure is. Agent Foundations asks what it means to reason from inside that structure.**

This is one of the most important transitions in the whole cycle.

Without O, the WORLD may be:

formal,

dynamical,

coherent,

and empirically measurable,

yet still described entirely from an external God's-eye perspective.

With O, the WORLD becomes **inhabited by a bounded observer**.

---

21. The Effective WORLD

We can now state the proposed operational definition.

> **Effective WORLD.**
> An Effective WORLD is a bounded structure containing effective distinctions, operational dynamics, compositional coherence, measurable realization, and an embedded observer relation.

Symbolically:

$\mathcal{W}$ = (V,F,C,M,O). (21.1)

The five coordinates answer:

V — **What can be distinguished?**

F — **What can happen?**

C — **What belongs together?**

M — **What is actually realized?**

O — **From where can the structure be known and acted upon?**

This is not proposed as a complete metaphysics.

It is a candidate **minimal operational definition of worldhood for a bounded intelligent system**.

---

22. Why the Fifth Step Does Not End the Story

If O completed the construction permanently, the sequence would be a line:

V → F → C → M → O. (22.1)

But embedded agency gives us a reason why the line should bend back toward its beginning.

An embedded observer is bounded.

Let:

K_env = effective complexity of the relevant environment,

K_O = representational capacity of the observer.

For a nontrivial embedded system:

$$K_O \ll K_{env}. \quad (22.2)$$

The observer therefore cannot maintain a complete micro-description of its environment.

It must compress.

It must coarse-grain.

It must decide which distinctions matter.

That is:

$$O \rightarrow \text{abstraction pressure}. \quad (22.3)$$

And therefore:

$$O \rightarrow V'. \quad (22.4)$$

The fifth research programme naturally regenerates the first problem.

---

23. Handoff V — Bounded Observers Must Abstract

The fifth productive handoff is:

$$\text{Agent Foundations} \rightarrow \text{Natural Abstraction}. \quad (23.1)$$

This is the edge that turns the sequence into a cycle.

The reasoning is simple.

Embeddedness implies boundedness.

Boundedness implies representational limitation.

Representational limitation implies compression pressure.

Compression pressure raises the question:

> **Which compressed distinctions preserve what matters?**

That is precisely the Natural Abstraction problem.

We may write:

$$\text{Embeddedness} \rightarrow \text{Boundedness} \rightarrow \text{Compression} \rightarrow \text{Abstraction}. \quad (23.2)$$

Thus:

$$O \rightarrow V'. \quad (23.3)$$

The loop closes.

---

24. The World-Bearing Loop

The complete research-complementarity cycle is now:

Natural Abstraction
→ Active Inference
→ Compositional World Modeling
→ Representation Geometry
→ Agent Foundations
→ Natural Abstraction. (24.1)

Its structural counterpart is:

Distinction → Consequence → Coherence → Realization → Perspective → Distinction. (24.2)

And its WORLD-building interpretation is:

$V \to (V,F) \to (V,F,C) \to (V,F,C,M) \to (V,F,C,M,O) \to V'$. (24.3)

Call this a:

World-Bearing Loop

A World-Bearing Loop is a closed sequence of complementary operations whose joint output supplies the structural requirements of an Effective WORLD and whose final embedded perspective regenerates the need for abstraction.

This is stronger than a taxonomy.

The research programmes do not merely occupy neighboring boxes.

Their strongest interfaces can be interpreted as **progressive structural completion**.

---

25. What Each Programme Adds

The cycle can be summarized compactly.

Research concentration	Dominant contribution	What is added to the WORLD
Natural Abstraction	non-arbitrary coarse-graining	V — distinctions
Active Inference	inference, action, generative dynamics	F — consequence
Compositional / Topos approaches	formal interfaces and coherence	C — composition
Representation Geometry / mechanistic measurement	measurable internal structure	M — realization
Agent Foundations / Embedded Agency	bounded self-referential standpoint	O — perspective

The critical idea is not:

> five programmes correspond one-to-one to five ontological substances.

It is:

> **each programme has developed particularly strong machinery around one structural incompleteness left salient by another.**

---

26. The Main Highlight Line

We can now state the article's central argument in its most compact form:

> **Abstractions need dynamics. Dynamics need a coherent frame. Formal frames need measurable realization. Realized structures need an embedded observer. Bounded observers need abstraction. Close those interfaces, and a collection of research programmes begins to look like a theory of how an Effective WORLD is formed.**

This is the first major conclusion.

But it is not yet the deepest one.

The loop currently explains:

how a WORLD can be constituted.

It does not yet explain:

how a WORLD can discover that it has become inadequate.

That requires history.

---

27. A World-Bearing Loop Is Not Necessarily a Temporal Pipeline

Before proceeding, an important caveat is necessary.

The sequence:

$$V \rightarrow F \rightarrow C \rightarrow M \rightarrow O \quad (27.1)$$

should not be interpreted as a literal chronology in which an intelligent system first learns abstractions, then later acquires dynamics, then later composition, and so forth.

In real systems:

M constrains V from the beginning.

O constrains F.

C may be required before some dynamics can even be defined.

F may determine which abstractions survive.

The research programmes overlap substantially.

Therefore:

$$\text{research handoff order} \neq \text{ontological construction order.} \quad (27.2)$$

The cycle is an **interface ordering**.

It says:

> when we emphasize one problem, another problem becomes especially visible at its boundary.

The true architecture is a network.

The ring highlights a particularly informative route through that network.

---

28. Why Overlap Strengthens Rather Than Weakens the Federation

One might object:

> Active Inference already contains representation learning.
> Agent Foundations also studies world models.
> Compositional methods can influence abstraction.
> Representation Geometry can study dynamics.

Correct.

That does not invalidate the architecture.

It prevents a simplistic one-to-one interpretation.

A research federation should contain overlapping jurisdiction.

Indeed, overlap is useful because it creates regions where two traditions can directly constrain each other.

The relevant question is therefore not:

Does programme A exclusively own V?

It is:

> Does programme A provide unusually powerful machinery for understanding some part of V and its interfaces?

That is a much weaker—and much more defensible—claim.

---

29. Incompleteness Is Not a Defect

This leads to an important methodological point.

Suppose Active Inference does not provide a complete theory of:

V,

C,

M,

and O.

That is not necessarily a weakness.

Its power may arise precisely because it concentrates formal machinery on inference and action.

Similarly:

Topos-style approaches need not provide a complete empirical account of neural geometry.

Representation Geometry need not solve reflective decision theory.

Agent Foundations need not supply a unique theory of natural abstraction.

Therefore:

> **incompleteness relative to the WORLD tuple is not evidence that a research programme has failed. It is what makes genuine complementarity possible.**

A useful federation should preserve specialization rather than erase it.

---

30. The First Closure: Synchronic WORLD Constitution

At a given stage n, suppose the five conditions are sufficiently satisfied:

V_n,

F_n,

C_n,

M_n,

O_n.

Then we have an Effective WORLD:

$\boldsymbol{\mathcal{W}}_n = (V_n, F_n, C_n, M_n, O_n)$. (30.1)

Call this **synchronic closure**.

It means:

> enough structure exists at one stage for the bounded observer to distinguish, predict, act, compose, measure, and inhabit an operational world.

This does not imply complete knowledge.

Nor does it imply that the WORLD is correct.

It means only that it is sufficiently closed to operate.

---

31. Workable Closure Is Not Final Truth

A WORLD can be coherent and wrong.

It can contain:

useful variables,

successful dynamics,

formal composition,

stable realization,

and an embedded observer,

yet still fail under new conditions.

Therefore:

Operational closure ≠ ontological finality. (31.1)

A good intelligence architecture requires something paradoxical:

enough Closure to operate,

but enough openness to revise.

This was the central problem of the foundational Five-Regime article.

We can now reintroduce it at a higher level.

---

32. The Missing Dimension Is Historical

The WORLD tuple:

$\mathcal{W}$ = (V,F,C,M,O) (32.1)

is largely synchronic.

It describes what the world contains at a given stage.

But persistent intelligence has history.

The observer acts.

The WORLD answers.

Some expectations succeed.

Others fail.

Some discrepancies are absorbed through ordinary updating.

Others persist.

Therefore every interaction should potentially produce:

an admitted trace T_t

and:

an unresolved residual r_t.

Write:

$(T_t, r_t) = \Pi \boldsymbol{\mathcal{W}}, P, O(x_t)$. (32.2)

The WORLD now begins to accumulate a past.

---

33. From Effective WORLD to Historical WORLD

Introduce two ledgers:

$L^+_{t+1} = L^+_t \oplus T_t$. (33.1)

$L^-_{t+1} = L^-_t \oplus r_t$. (33.2)

Then define:

$\boldsymbol{\mathcal{W}}^H = (\boldsymbol{\mathcal{W}}, L^+, L^-)$. (33.3)

Call $\boldsymbol{\mathcal{W}}^H$ a:

Historical WORLD

An Effective WORLD can operate.

A Historical WORLD additionally remembers:

what it successfully admitted,

and:

what it repeatedly failed to absorb.

This distinction is essential.

Without L^+, the system lacks accountable continuity.

Without L^-, the system risks becoming self-sealing.

---

34. Why the Residual Ledger Matters to the Research Federation

The residual can be interpreted differently by each research tradition.

Natural Abstraction may see:

a coarse-graining that no longer preserves relevant structure.

Active Inference may see:

persistent mismatch under the current generative model.

Compositional approaches may see:

failure of the present frame or interfaces to compose adequately.

Representation Geometry may see:

trajectories leaving a stable basin or internal structure reorganizing.

Agent Foundations may see:

an embedded agent whose self/world model no longer supports adequate reasoning.

SMFT describes the same candidate event as:

residual accumulation under the current Declaration.

The earlier comparison already suggested almost exactly this multi-framework translation. 探討 SMFT 串聯其它名門正派成為五行流轉的可行性

These descriptions are not automatically equivalent.

But they may sometimes be **projections of the same structural failure event**.

That possibility deserves formal treatment.

---

35. From Historical WORLD to Self-Revising WORLD

A Historical WORLD can remember failure.

But it cannot yet necessarily change the structure that generated the failure.

Introduce a revision operator:

U. (35.1)

Then:

$\mathcal{W}_{n+1} = U(\mathcal{W}_n, L^+_n, L^-_n; P)$. (35.2)

Define:

$\mathcal{W}^R = (\mathcal{W}, L^+, L^-, U; P)$. (35.3)

Call this a:

Self-Revising WORLD

The semicolon before P is deliberate.

Purpose conditions revision without being treated as just another coordinate of worldhood.

The system now possesses:

a world,

a history,

a remainder,

and a mechanism by which that history can alter the world itself.

---

36. Three Levels of WORLD

The architecture therefore distinguishes:

W0 — Effective WORLD

$\mathcal{W}$ = (V,F,C,M,O). (36.1)

It can operate.

W1 — Historical WORLD

$\mathcal{W}^{H}$ = ($\mathcal{W}$,L^{+},L^{-}). (36.2)

It can remember both success and unresolved failure.

W2 — Self-Revising WORLD

$\mathcal{W}^{R}$ = ($\mathcal{W}$,L^{+},L^{-},U;P). (36.3)

It can revise its own effective world structure.

This hierarchy provides a much clearer position for SMFT.

---

37. Where SMFT Lives

SMFT does not need to claim:

Natural Abstraction is part of SMFT.

Active Inference is part of SMFT.

Topos is part of SMFT.

Representation Geometry is part of SMFT.

Agent Foundations is part of SMFT.

That would be unnecessarily imperial.

A cleaner positioning is:

> **The five external research concentrations primarily illuminate how an Effective WORLD may be constituted. SMFT proposes a candidate grammar for how such a WORLD becomes historically accountable and revisable.**

Graphically:

(V,F,C,M,O)

↓
Trace / Residual
↓
Dual Ledger
↓
Latching / Revision
↓
(V′,F′,C′,M′,O′). (37.1)

SMFT therefore operates mainly along the **vertical temporal axis**.

The five-programme federation operates mainly along the **horizontal constitutive axis**.

This is a much more precise relationship.

---

38. Horizontal and Vertical Completeness

We can now define two kinds of incompleteness.

Horizontal incompleteness

One or more coordinates of:

(V,F,C,M,O)

are insufficiently developed.

The WORLD cannot fully support:

distinction,

consequence,

coherence,

realization,

or embedded perspective.

Vertical incompleteness

The WORLD operates now but cannot:

preserve trace,

retain unresolved residual,

judge structural inadequacy,

or revise itself.

The first is a problem of:

world constitution.

The second is a problem of:

world persistence and revision.

A mature theory of persistent intelligence may require both.

---

39. Constitution + Revision

The whole architecture can now be compressed into two equations.

WORLD Constitution

$$V \rightarrow (V,F) \rightarrow (V,F,C) \rightarrow (V,F,C,M) \rightarrow (V,F,C,M,O)=\mathcal{W}. \quad (39.1)$$

WORLD Revision

$$\mathcal{W}_n \rightarrow (T_n,r_n) \rightarrow (L^+_n,L^-_n) \rightarrow U \rightarrow \mathcal{W}_{n+1}. \quad (39.2)$$

Together:

WORLD Constitution + WORLD Revision = Recursive World Formation. (39.3)

This, rather than a five-school classification, is the central architecture proposed in this article.

40. One Structural Failure, Five Framework Descriptions

The research-federation proposal becomes more useful if it can do more than place programmes on a map.

A stronger test is:

> **Can several frameworks describe the same failure event from different structural projections?**

Consider a system whose current WORLD:

$$\mathcal{W}_n = (V_n,F_n,C_n,M_n,O_n) \quad (40.1)$$

has become inadequate.

The external phenomenon is one event:

E* = persistent WORLD-model inadequacy. (40.2)

Different programmes may nevertheless describe E* differently.

Natural Abstraction might say:

> the current coarse-graining no longer preserves the variables required for prediction or control.

Active Inference might say:

> persistent mismatch remains under the current generative model despite ordinary

belief and policy updating.

Compositional world modeling might say:

> the current frame no longer supports coherent composition of the relevant local models.

Representation Geometry might say:

> the system's trajectory no longer remains within the previously stable representational organization.

Agent Foundations might say:

> the embedded agent's current self/world representation no longer supports adequate reasoning under its bounded information and computational constraints.

SMFT might say:

> persistent structured residual has exceeded what the current Declaration can absorb, weakening the current latch and creating pressure for re-declaration.

The earlier comparative work already suggested almost exactly this multi-framework translation: one event can appear as an inadequate frame, unresolved uncertainty, failing coarse-graining, persistent variational mismatch, representational-basin failure, or residual-driven re-declaration depending on the descriptive layer. 探討 SMFT 串聯其它名門正派成為五行流轉的可行性

The important possibility is therefore:

> **different theoretical vocabularies may sometimes be projections of one underlying transition rather than mutually exclusive explanations.**

---

41. The Cross-Framework Event

Define a **Cross-Framework Event** as an event whose structural role can be represented within several theoretical languages.

Let:

E_A

denote the description of event E under Framework A.

Let:

E_B

denote the description under Framework B.

A translation map is then:

$T_{A\to B} : E_A \to E_B.$ (41.1)

However, the translation should not normally be assumed exact.

Instead:

$T_{A\to B}(E_A) \approx E_B$. (41.2)

The approximation symbol matters.

Each framework may:

preserve some structure,

discard some structure,

introduce different primitives,

or distinguish cases that another framework merges.

Therefore define a **translation residual**:

$\varepsilon_{A\to B} = \mathrm{Loss}[T_{A\to B}(E_A), E_B]$. (41.3)

A good cross-framework interface has:

$\varepsilon_{A\to B}$ small enough to preserve the structure relevant to the shared question. (41.4)

This provides a more rigorous meaning for “linking schools.”

---

42. Translation Is Not Reduction

Suppose Active Inference and SMFT both describe a persistent mismatch.

That does not imply:

prediction error = residual. (42.1)

Likewise:

representational basin = Declaration, (42.2)

or:

formal frame = WORLD, (42.3)

would be too strong.

The correct relation is more like:

prediction error may contribute to residual evidence;

a representational basin may instantiate some aspects of a latched WORLD;

a formal frame may constrain some components of Declaration.

Thus:

Framework A concept ≠ Framework B concept. (42.4)

But:

Framework A concept ↔ partially overlapping structural role. (42.5)

The purpose of the federation is therefore not synonym creation.

It is **interface engineering**.

---

43. What a Useful Translation Must Preserve

For a translation to be scientifically useful, it should preserve at least one invariant.

Suppose:

$I_A(E_A)$

is an invariant or measurable property in Framework A.

A meaningful translation should approximately satisfy:

$$I_B[T_{A\to B}(E_A)] \approx I_A(E_A). \quad (43.1)$$

Examples might include:

transition ordering,

stability class,

intervention response,

information retained,

identity continuity,

or failure/recovery behavior.

Without such preservation, the translation is merely linguistic resemblance.

Thus a future research federation should ask:

> **What is preserved when a problem is translated from one framework to another?**

This may be more important than asking whether the frameworks use similar terminology.

---

44. The Federation as a Network of Partial Translations

Let the five anchor programmes be:

A_1 = Natural Abstraction

A_2 = Active Inference

A_3 = Compositional / Topos approaches

A_4 = Representation Geometry / Mechanistic Measurement

A_5 = Agent Foundations.

Then define a network:

$\mathfrak{F} = (A,T), \quad (44.1)$

where:

$A = \{A_1,\ldots,A_5\} \quad (44.2)$

and:

$T = \{T_{ij}\}. \quad (44.3)$

Each T_{ij} is a partial translation or handoff.

The World-Bearing Loop corresponds to a particularly important cycle:

$A_1 \to A_2 \to A_3 \to A_4 \to A_5 \to A_1. \quad (44.4)$

But the full federation contains many additional edges.

For example:

Natural Abstraction ↔ Representation Geometry

is important because abstractions must be measurable.

Active Inference ↔ Agent Foundations

is important because inference is performed by bounded agents.

Topos ↔ Natural Abstraction

is important because abstractions must compose.

Thus the true federation is a network.

The five-cycle is a **productive backbone**.

---

45. Productive Closure

This distinction allows us to define the central article concept more carefully.

A sequence:

$A_1 \to A_2 \to \ldots \to A_n \to A_1 \quad (45.1)$

has **productive closure** when:

1. every handoff adds a nontrivial structural capability;
2. no step is merely terminological relabeling;
3. the final step regenerates a genuine problem addressed by the first;
4. the closed sequence constructs a richer object than any node alone.

In the present proposal:

Natural Abstraction adds V.

Active Inference adds F.

Compositional modeling adds C.

Representation measurement adds M.

Agent Foundations adds O.

The result is:

$\mathcal{W} = (V,F,C,M,O).$ (45.2)

Then bounded O regenerates the need for improved V.

Therefore the cycle closes:

$O \rightarrow V'.$ (45.3)

This is the proposed productive closure.

---

46. Why This Is More Than Interdisciplinary Collaboration

Ordinary interdisciplinary work often takes the form:

Theory A + Theory B.

The World-Bearing Loop proposes something stronger:

Theory A produces a structure whose unresolved boundary becomes the entry point of B.

B produces another structure whose boundary becomes the entry point of C.

Eventually the cycle returns to A with a transformed problem.

Thus:

$A_1 \rightarrow A_2 \rightarrow \ldots \rightarrow A_n \rightarrow A_1'.$ (46.1)

The prime is important.

The final A_1' is not the original A_1.

It operates on a WORLD already altered by the complete cycle.

Therefore the federation is recursive:

$\mathcal{W}_n \rightarrow \mathcal{W}_{n+1}.$ (46.2)

The research relation itself begins to resemble the world-revision process it studies.

---

47. The World-Bearing Loop Is a Spiral, Not a Circle

Suppose the first pass produces:

$\mathcal{W}_0.$

The embedded observer O_0 then discovers limitations.

That produces:

$V_1 \neq V_0$. (47.1)

The second pass generates:

F_1,

C_1,

M_1,

O_1.

Thus:

$\boldsymbol{\mathcal{W}}_1 \neq \boldsymbol{\mathcal{W}}_0$. (47.2)

The research cycle becomes:

$$V_0 \rightarrow F_0 \rightarrow C_0 \rightarrow M_0 \rightarrow O_0$$
$$\downarrow$$
$$V_1 \rightarrow F_1 \rightarrow C_1 \rightarrow M_1 \rightarrow O_1$$
$$\downarrow$$
$$V_2 \rightarrow \ldots \quad (47.3)$$

This is not simple repetition.

It is **recursive WORLD construction**.

The structural analogy with the Five-Regime article is now clear:

both cycles are better represented as spirals whose previous outputs become the boundary conditions of the next turn.

---

48. The Five-Regime Runtime Can Now Be Overlaid

The foundational article proposed runtime modes:

$q \in \{G, A, C_r, S, R\}$. (48.1)

where:

G = Generation

A = Activation

C_r = Closure

S = Selection

R = Retention.

These modes act upon the WORLD:

$q(t) : \boldsymbol{\mathcal{W}} \rightarrow \boldsymbol{\mathcal{W}}'$. (48.2)

The WORLD coordinates:

(V,F,C,M,O)

describe what structure currently exists.

The Five-Regime modes describe what transformation currently dominates.

This gives a useful two-layer architecture:

WORLD coordinates = configuration.

Five-Regime modes = runtime.

---

49. Approximate Phase Alignment

Although the two decompositions are not identical, an approximate alignment appears:

Natural Abstraction / V formation
↔ Generation

Active Inference / F
↔ Activation

Compositional WORLD structure / C
↔ Closure

Representation measurement / M
↔ Selection

Embedded bounded perspective / O
↔ Retention and renewed Generation.

Schematically:

$$V \rightarrow F \rightarrow C \rightarrow M \rightarrow O \rightarrow V' \quad (49.1)$$

and:

$$G \rightarrow A \rightarrow C_r \rightarrow S \rightarrow R \rightarrow G'. \quad (49.2)$$

The correspondence is suggestive.

But it should remain:

[A] structural analogy,

not:

[D] derivation.

The two cycles were constructed for different purposes.

Their approximate phase alignment should therefore be treated as a **testable structural coincidence** rather than an identity.

---

50. Why the Non-Identity Matters

If the two cycles were declared identical, several distortions would immediately appear.

Agent Foundations is not merely Retention.

Representation Geometry is not merely Selection.

Active Inference is not merely Activation.

Likewise the Closure regime C_r is not identical to the compositional coordinate C.

The correct picture is:

research programmes constrain WORLD coordinates;

runtime regimes transform WORLD coordinates.

Thus:

School Map × Runtime Map (50.1)

forms a two-dimensional architecture.

This is more powerful than either map alone.

---

51. A School–Regime Matrix

Let:

W_{ij} (51.1)

measure how strongly research programme i informs runtime regime j.

Then:

$W \in \mathbb{R}^{5\times5}$ (51.2)

for the five anchor programmes and five runtime regimes.

The matrix should not be diagonal.

Natural Abstraction may strongly inform:

G,

S,

and C_r.

Active Inference may inform:

A,

S,

and R.

Compositional approaches may strongly inform:

C_r,

but also G and S.

Representation Geometry may provide measurements across all five.

Agent Foundations may constrain all five because the controller itself is embedded.

Therefore a non-diagonal W is expected.

That is not a failure of the fivefold architecture.

It is evidence that the two decompositions capture different dimensions.

---

52. Measurement and Embeddedness as Cross-Cutting Axes

This explains why the earlier planning suggested a 5+2 representation.

The five runtime functions are:

G,A,C_r,S,R.

But two research concerns cut across the entire runtime:

M* = measurement / observability

E* = embeddedness / self-reference.

Representation Geometry is concentrated around M*.

Agent Foundations is concentrated around E*.

Thus a research programme may be positioned by:

p = (g,a,c,s,r,m,e). (52.1)

This seven-dimensional positioning space is optional.

The simpler WORLD tuple remains sufficient for the main article.

But the 5+2 representation may be useful in later quantitative mapping.

---

53. A Stronger Interpretation of Agent Foundations

Agent Foundations may occupy a particularly important position.

It is not simply one more specialist in the ring.

It asks:

> **Who controls the controller when the controller is inside the WORLD it is revising?**

This is a meta-level problem.

The Five-Regime framework assumes that a system can decide:

when to generate,

when to act,

when to close,

when to select,

when to retain,

when to revise.

But the entity making those decisions is itself:

bounded,

self-referential,

and embedded.

Therefore Agent Foundations naturally constrains the meta-controller:

$Q : (q, \mathcal{W}, L^+, L^-, P) \to q'. \quad (53.1)$

This makes Embedded Agency relevant across the whole cycle.

The earlier source explicitly emphasizes precisely this difficulty: the agent is smaller than its environment and lacks a clean external boundary from which to reason. 探討 SMFT 串聯其它名門正派成為五行流轉的可行性

---

54. A Stronger Interpretation of Representation Geometry

Representation Geometry has a similarly cross-cutting role.

Its deepest contribution may not be:

Selection.

It may be:

instrumentation of the entire runtime.

If Five-Regime theory is correct, one might seek distinct measurable signatures for:

Generation,

Activation,

Closure,

Selection,

Retention,

and transitions between them.

Thus:

M_q : internal system → regime evidence. (54.1)

The empirical programme becomes:

infer $q(t)$

from measurable geometry,

then intervene,

and test whether altering the inferred regime changes behavior in the predicted direction.

This is substantially stronger than post hoc visualization.

---

55. A Research Federation Needs Interface Experiments

The federation becomes scientifically valuable only when an interface can be tested.

The earlier comparative work already recommended moving from abstract “unification” claims toward precise shared open problems and collaborative experiments. 探討 SMFT 串聯其它名門正派成為五行流轉的可行性

The present architecture suggests several such experiments.

---

56. Interface Experiment I — Natural Abstraction × Active Inference

Question

Do more natural abstractions improve inference and control across distribution shift?

Construct several representations:

$V_1, V_2, \ldots, V_k$.

Measure:

predictive performance,

control efficiency,

transfer,

robustness,

and representation stability.

Prediction:

representations with stronger cross-context abstraction properties should support more robust operative dynamics.

Formally:

$$\text{Naturalness}(V) \uparrow \Rightarrow \text{Transfer}(F|V) \uparrow \quad (56.1)$$

as a hypothesis.

This experiment directly tests the V→F handoff.

---

57. Interface Experiment II — Active Inference × Compositional WORLD Modeling

Question

When should persistent within-model failure trigger frame restructuring rather than further inference inside the current model?

Let:

D_n

denote the current generative/formal frame.

Ordinary updating gives:

$$x_t \rightarrow x_{t+1} \mid D_n. \quad (57.1)$$

Frame revision gives:

$$D_n \rightarrow D_{n+1}. \quad (57.2)$$

The experiment should distinguish environments containing:

parameter shifts

from:

structural shifts.

A successful architecture should not reframe unnecessarily under parameter change, yet should eventually reframe under structural inadequacy.

This is precisely the boundary between:

within-WORLD inference

and:

WORLD reconstruction.

---

58. Interface Experiment III — Compositional

Structure × Representation Geometry

Question

Do formally predicted changes of frame correspond to measurable representational reorganizations?

Suppose a formal model predicts a transition:

$C_1 \rightarrow C_2$. (58.1)

Seek corresponding empirical signatures:

$M_1 \rightarrow M_2$. (58.2)

Candidate signatures include:

subspace rotation,

basin transition,

connectivity reorganization,

dimensionality shift,

or hysteresis.

The important test is intervention.

If C_1 and C_2 are more than analyst-imposed descriptions, manipulations predicted to stabilize one frame should alter measurable dynamics accordingly.

Thus:

formal frame change → measurable structural change. (58.3)

This is the C→M interface.

---

59. Interface Experiment IV — Representation Geometry × Embedded Agency

Question

Which externally measurable representations are actually available to the agent's own decision process?

Suppose a researcher identifies latent structure:

$Z_{external}$. (59.1)

Does the agent possess access to it?

Define:

Accessibility(Z) = effect of Z on the agent's own counterfactual reasoning, decisions, or self-

model. (59.2)

Then:

measurable ≠ internally accessible. (59.3)

This distinction may expose an important gap between:

representation analysis

and:

embedded cognition.

A representation can exist in the machine without existing as an explicit or usable variable for the agent.

---

60. Interface Experiment V — Embedded Agency × Natural Abstraction

Question

Does increased resource boundedness systematically change which abstractions become useful or stable?

Vary:

memory capacity,

compute,

observation bandwidth,

planning depth.

Then infer:

V(O_budget). (60.1)

The hypothesis is:

observer constraints alter the optimal coarse-graining.

Thus:

∂V/∂Budget ≠ 0. (60.2)

Yet some abstractions may remain stable across a wide range of observers.

Those would be especially strong candidates for natural abstractions.

This experiment directly closes the research cycle.

---

61. The Cycle Produces a New Research Object

The five programmes are usually studied separately.

The federation proposal produces a new object:

the interface itself

Examples include:

natural abstraction ↔ inference,

inference ↔ frame composition,

formal frame ↔ measurable realization,

measurement ↔ embedded accessibility,

embeddedness ↔ abstraction.

These interfaces may contain research questions that are not central to either neighboring programme alone.

Thus the federation does not merely organize existing work.

It may generate new work.

---

62. A Research Gap Can Be Defined as an Unresolved Handoff

Let H_ij denote the interface between programmes i and j.

Define:

Gap(H_ij) = unresolved structure required to translate useful outputs of i into usable inputs for j. (62.1)

Then a research roadmap can be built around:

minimize Gap(H_ij). (62.2)

The most valuable new work may therefore occur at the boundaries.

This reverses the usual institutional logic.

Instead of asking:

> Which school should win?

ask:

> **Which handoff is currently weakest?**

That question is often much more actionable.

---

63. The WORLD Can Fail at Different Interfaces

The WORLD tuple also allows failure localization.

V-failure

The available variables are wrong or insufficient.

F-failure

The operative dynamics are inadequate.

C-failure

Local models cannot be composed coherently.

M-failure

The proposed structure lacks reliable realization.

O-failure

The observer cannot access or reason with the required structure.

These failures need different repairs.

Thus:

Residual → identify failing coordinate → choose repair depth. (63.1)

This creates a direct link to the Five-Regime repair hierarchy.

---

64. WORLD Revision Need Not Rewrite Everything

Suppose failure occurs primarily in V.

Then:

$V \to V'$ (64.1)

may suffice while much of:

F,C,M,O

is preserved.

Likewise a failure in C may require:

$C \to C'$ (64.2)

without discarding all abstractions.

Thus WORLD revision should be modular:

$\mathcal{W}' = U_k(\mathcal{W}, L^-)$, (64.3)

where k identifies the component or interface requiring revision.

This prevents the theory from equating every structural failure with total WORLD replacement.

A good system revises the shallowest adequate level.

---

65. Cross-School Research Can Help Diagnose Revision Depth

This provides another reason the federation matters.

A persistent mismatch may appear to one framework as:

“prediction failure.”

But another framework may diagnose:

“the abstraction itself is wrong.”

A third may diagnose:

“the problem is compositional.”

A fourth may show:

“the supposed transition is not actually present in the representation.”

A fifth may reveal:

“the representation exists externally but is inaccessible to the embedded agent.”

Therefore multiple frameworks can function as **differential diagnosis**.

Their disagreement can be informative.

---

66. Agreement and Disagreement Are Both Useful

If several frameworks independently indicate the same revision:

Evidence_revision ↑. (66.1)

But disagreement can reveal a translation error.

For example:

Active Inference predicts model inadequacy,

while Representation Geometry shows no corresponding structural transition.

Possible explanations include:

1. the geometric measurement is insufficient;
2. the active-inference description is too coarse;
3. the same behavior can be realized without the proposed structural transition;
4. the two theories are describing different levels.

Thus:

Cross-framework disagreement → residual for the federation itself. (66.2)

This is an important principle.

The research federation should itself be self-revising.

---

67. The Federation Has Its Own Dual Ledger

This idea can be pushed one step further.

Let:

L^+_F

record successful cross-framework translations.

Let:

L^-_F

record unresolved mismatches.

Then:

$L^+_F \leftarrow$ confirmed interfaces. (67.1)

$L^-_F \leftarrow$ failed translations / contradictory predictions. (67.2)

If L^-_F accumulates strongly around some interface, the federation map should be revised.

Thus the methodological architecture mirrors the object-level theory:

Research federation
→ translation
→ success/residual
→ ledger
→ interface revision. (67.3)

This is not necessary for the main framework, but it is a natural consequence of taking residual seriously.

---

68. What SMFT Contributes to the Federation

The strongest defensible positioning is not:

> SMFT unifies all five programmes.

It is:

> **SMFT proposes a candidate grammar for the transitions among WORLD constitution, observation, residual accumulation, latching, and WORLD revision.**

Its characteristic objects are:

Declaration,

Purpose,

Observer,

Gate,

Trace,

Filtration,

Residual,

Latching,

Revision.

The previous comparative work explicitly proposed using this Core as the shared collaboration layer while keeping mathematical extensions and traditional interpretations separate. 探討 SMFT 串聯其它名門正派成為五行流轉的可行性

Thus SMFT's candidate contribution is largely:

diachronic coordination.

---

69. What SMFT Does Not Need to Claim

SMFT need not claim that it supplies:

the final theory of abstraction;

the final theory of generative inference;

the final formal semantics of composition;

the final empirical theory of representation;

or:

the final theory of embedded agency.

Indeed, making those claims would weaken the federation.

A stronger role is:

> **a grammar of when a WORLD remains adequate, when residual indicates inadequacy, what level should be revised, and how historically accountable revision can occur.**

This position leaves substantial mathematical work to the programmes already strongest in their respective areas.

---

70. Research Federation as Division of Mathematical Labor

The federation can therefore be interpreted as a division of labor.

Natural Abstraction develops the mathematics of:

compression and stable macrovariables.

Active Inference develops the mathematics of:

generative inference, prediction, and action.

Compositional approaches develop the mathematics of:

interfaces, composition, and formal frames.

Representation Geometry develops the mathematics and measurement of:

internal organization and transitions.

Agent Foundations develops the logic of:

embedded reasoning, self-reference, and bounded agency.

SMFT proposes:

historical residual governance and world revision.

No one programme needs to become all the others.

The gain comes from **composable specialization**.

---

71. From Division of Labor to WORLD Formation

We can now understand why your proposed highlight line is so important.

The article does not merely say:

five schools are complementary.

It says:

> **their complementarities can be ordered into constructive progress.**

The progression is:

abstraction
→ dynamics
→ coherence
→ realization
→ perspective.

At the end of this progression we obtain:

$\mathcal{W} = (V,F,C,M,O).$ (71.1)

The final bounded perspective then generates renewed abstraction pressure:

$O \to V'.$ (71.2)

So:

the supplement chain closes.

And closure produces something new:

an Effective WORLD.

---

72. The Strongest Version of the Highlight

The most compact statement may be:

> **Natural Abstraction gives the system something to think with. Active Inference gives those distinctions consequences. Compositional modeling makes the consequences parts of one coherent world. Representation Geometry asks whether that world is actually realized. Agent Foundations puts the thinker inside the realized world. Once inside, the thinker is bounded and must abstract again.**

Then:

> **The cycle does not merely connect theories. It progressively constructs the conditions of worldhood.**

This should probably be one of the highlighted passages in the final article.

---

73. A Second Highlight: WORLD Is an Emergent Product of Interfaces

No single research programme in the federation equals:

$\mathcal{W}$.

Instead:

$\mathcal{W}$ emerges from the closure of interfaces.

This gives:

WORLD ≠ Σ isolated theories. (73.1)

More accurately:

WORLD ≈ Closure(complementary interfaces). (73.2)

The difference is important.

Simply collecting theories does not produce a world.

The outputs must compose.

Thus the fundamental object is:

the closed interface architecture.

---

74. A Third Highlight: Self-Revising WORLD Requires a Second Closure

The first closure is:

V,F,C,M,O → $\mathcal{W}$. (74.1)

The second closure is:

$\mathcal{W}$ → Trace/Residual → Ledger → Revision → $\mathcal{W}'$. (74.2)

Call them:

Constitutive Closure

and:

Recursive Closure.

Then:

Constitutive Closure + Recursive Closure = Persistent World Formation. (74.3)

This two-closure structure may become the companion paper's principal theoretical contribution.

---

75. Constitutive Closure

Constitutive Closure asks:

> Is there enough structure here for an embedded observer to possess an operational world?

It requires at least:

distinction,

dynamics,

composition,

realization,

perspective.

Thus:

$C_{const}(\mathcal{W}) \geq \theta_{const}$. (75.1)

The exact metric remains unspecified.

But conceptually:

below threshold,

the pieces do not yet form a workable WORLD.

76. Recursive Closure

Recursive Closure asks:

> Is the WORLD capable of preserving the consequences of its own operation strongly enough to revise itself without losing all continuity?

It requires:

Trace,

Residual,

Ledger,

Revision.

Thus:

$C_{rec}(\mathcal{W}^R) \geq \theta_{rec}$. (76.1)

A system can possess strong constitutive closure but weak recursive closure.

It can operate effectively yet be unable to change its WORLD when the WORLD becomes inadequate.

That distinction seems highly relevant to current AI.

---

77. Intelligence May Require Both Closures

This suggests a broader conjecture:

> **Persistent intelligence requires both constitutive closure and recursive closure.**

Constitutive closure allows the system to:

operate inside a world.

Recursive closure allows it to:

change the world without losing itself.

We may write:

Persistent Intelligence ≈ Constitutive WORLD + Governed WORLD Revision. (77.1)

This is not intended as a formal definition of all intelligence.

It is a structural hypothesis about **persistent self-revising intelligence**.

---

78. Why This Goes Beyond a World Model

Modern AI commonly uses the phrase:

world model.

But the WORLD defined here is richer.

A conventional world model may be primarily predictive:

M_pred : history → future prediction. (78.1)

An Effective WORLD additionally requires:

V,

F,

C,

M,

O.

A Self-Revising WORLD additionally requires:

L^+,

L^-,

U,

conditioned by P.

Thus:

predictive world model ⊂ possible component of Effective WORLD. (78.2)

The research federation is therefore aimed at a broader problem:

> not merely predicting the environment, but constituting the operational frame in which prediction, action, evidence, and revision are meaningful.

---

79. The WORLD Is Observer-Compatible, Not Observer-Arbitrary

Because O is part of $\mathcal{W}$, one might worry that the WORLD becomes purely subjective.

That does not follow.

The V, F, C, and M coordinates constrain what O can successfully sustain.

An observer cannot arbitrarily declare any WORLD and expect:

prediction,

action,

composition,

and measurement

to remain coherent.

Thus WORLD formation contains a tension between:

observer dependence

and:

external constraint.

One may write schematically:

$\mathcal{W}$ = Consistency(V,F,C,M,O). (79.1)

WORLD is therefore neither:

a completely observer-independent God's-eye object

nor:

a free subjective invention.

It is an **observer-compatible constrained construction**.

---

80. This Is Where Natural Abstraction Becomes Crucial Again

If multiple bounded observers independently converge on similar V under similar constraints, that provides evidence that the WORLD's distinctions are not arbitrary.

Thus Natural Abstraction can contribute a form of cross-observer stability.

Let observers $O_1,\ldots,O_n$ derive abstractions:

$V_1,\ldots,V_n$.

A candidate natural abstraction has:

Agreement($V_1,\ldots,V_n$) high (80.1)

despite:

$O_i \neq O_j$. (80.2)

This may provide one bridge between:

observer dependence

and:

structural objectivity.

The cycle therefore returns to its beginning at a deeper level.

---

81. The Federation as a Theory of Objectivity

This suggests an unexpected implication.

Objectivity need not require an observer-free description.

It may instead arise from:

different bounded observers

using different procedures

converging on structures that survive:

translation,

intervention,

composition,

and measurement.

Thus:

Objectivity ≈ cross-observer invariant structure. (81.1)

Natural Abstraction asks whether variables converge.

Active Inference asks whether predictions and actions work.

Compositional approaches ask whether structures compose.

Representation Geometry asks whether realization persists.

Agent Foundations asks whether bounded agents can access the structure.

SMFT asks whether the structure survives historical revision.

Together, these provide multiple filters on objectivity.

This is a deeper philosophical consequence of the federation.

---

82. A Candidate Research-Federation Invariant

Suppose some structure X appears in several frameworks.

Call X robust if:

$T_{A \to B}(X_A) \approx X_B$ (82.1)

for multiple framework pairs,

and:

X survives intervention and WORLD revision.

Then define schematically:

Robustness(X) = Translation Stability × Intervention Stability × Revision Stability. (82.2)

This is only a conceptual measure.

But it suggests how a federation might produce stronger evidence than any framework alone.

A structure is increasingly credible when it survives multiple independent representations.

---

83. The Federation Therefore Does More Than Divide Labor

The federation has three possible functions.

First — Supplementation

One programme adds structure another leaves implicit.

Second — Cross-validation

Different formalisms test whether the same structural event survives translation.

Third — Residual generation

Disagreement reveals where the current federation map is inadequate.

This is why the federation itself can become a research engine.

---

84. A WORLD Can Be Closed Without Being Correct

This caution deserves repetition.

Constitutive closure means:

the system possesses a sufficiently coherent operational world.

It does not mean:

the world corresponds perfectly to reality.

Indeed, a dangerous system may be very well closed.

Its V,F,C,M,O may reinforce one another strongly.

That can produce:

high internal coherence

with:

poor external adequacy.

Therefore a persistent intelligent system requires not only closure but **residual sensitivity**.

Without residual:

closure becomes dogmatism.

This connects directly to the Five-Regime article's distinction between healthy Closure and Closure lock.

---

85. Research Schools Can Also Become Closed in This Pathological Sense

The same pathology can affect theories.

A research tradition can develop:

its own variables,

its own methods,

its own benchmarks,

its own explanatory vocabulary,

and its own success criteria.

It may then become difficult for evidence from another framework to enter.

In our terminology:

the school has high internal C

but poor cross-framework permeability.

A research federation can counter this by maintaining translation interfaces.

Thus the article's architecture applies not only to AI systems but, cautiously, to the organization of research itself.

---

86. The Federation as an Anti-Lock Architecture

A federation can preserve diversity.

If one framework over-dominates, others can supply counterevidence.

For example:

formal elegance can be challenged by measurement;

measurement can be challenged by embedded-access questions;

embedded-agent models can be challenged by abstraction naturalness;

abstractions can be challenged by operational failure;

operational models can be challenged by compositional inconsistency.

Thus the research cycle has a built-in anti-lock property.

This is analogous—not identical—to the regulatory cycle in the Five-Regime framework.

---

87. The Wider Research Ecosystem

The five anchor programmes do not exhaust AI foundations.

Several adjacent fields naturally enter the architecture.

Causal Representation Learning

Strongly intersects:

V,F,C.

It asks which learned representations support causal and transferable structure.

Reinforcement Learning and Model-Based Control

Strongly intersect:

F

and runtime Activation/Selection.

Continual Learning

Strongly intersects:

Retention,

history,

and revision.

Meta-Learning

Concerns adaptation of learning procedures themselves and therefore touches:

G,

S,

and U.

Mechanistic Interpretability

Provides empirical machinery for M.

Predictive Processing

Overlaps substantially with Active-Inference-style world dynamics.

Dynamical Systems and Control Theory

Provide mathematics for:

regime switching,

stability,

hysteresis,

and latching.

These fields should be treated as a surrounding ecosystem rather than forced into additional fixed positions.

---

88. Why the Five Anchors Are Still Useful

If many other fields overlap, why retain the five anchors?

Because they provide a particularly clean route through five distinct structural questions:

1. What distinctions are natural?
2. What consequences follow?
3. What makes the pieces one world?
4. Where is the structure realized?
5. How can a bounded observer inhabit it?

The five anchors are therefore best treated as:

> **representative concentrations of expertise around five WORLD requirements.**

They are not canonical owners.

Future versions of the map should allow other programmes to replace, supplement, or split them.

---

89. The Federation Should Be Empirically Revisable

Let the current positioning map be:

$$\text{Map}_0. \quad (89.1)$$

New evidence may show that:

one programme spans more coordinates than expected;

one handoff is unnecessary;

a sixth WORLD coordinate is required;

two coordinates can be merged;

or:

the proposed cycle does not close.

Then:

$$\text{Map}_0 \rightarrow \text{Map}_1. \quad (89.2)$$

The federation should therefore obey the same principle as the theory:

> preserve residual and revise the map when necessary.

That methodological symmetry is important.

---

90. The Strongest Falsifier of the World-Bearing Loop

The most serious challenge would be to show that the five handoffs are rhetorically constructed rather than structurally necessary.

For example, suppose:

Active Inference alone already provides a sufficiently complete treatment of:

V,F,C,M,O.

Then the claimed complementarity cycle becomes much less important.

Or suppose:

Agent Foundations does not meaningfully regenerate the abstraction problem.

Then the loop fails to close.

Or suppose:

M is not a constitutive WORLD coordinate but only an external research convenience.

Then the WORLD tuple requires revision.

These are legitimate criticisms.

The article should invite them.

---

91. The Strongest Positive Evidence

Conversely, the World-Bearing Loop gains credibility if:

1. each handoff identifies a reproducible unresolved interface;
2. neighboring frameworks provide useful solutions to those interfaces;
3. the integrated architecture predicts failures not visible from one framework alone;
4. cross-framework translations preserve measurable invariants;
5. the closed architecture improves adaptation under WORLD change.

The strongest result would not be philosophical agreement.

It would be:

better explanatory and experimental performance through interface composition.

---

92. A Concrete Collaborative Benchmark

A future benchmark could require an agent to operate in an environment containing both:

ordinary parameter changes

and:

structural ontology changes.

The benchmark would evaluate:

Abstraction

Can the agent discover useful V?

Dynamics

Can it act and predict under F?

Composition

Can it combine local models under C?

Realization

Can internal signatures of these structures be identified under M?

Embeddedness

Can the agent reason about its own limitations under O?

Revision

Can persistent residual trigger selective updates:

$\boldsymbol{W}_n \rightarrow \boldsymbol{W}_{n+1}$?

Such a benchmark would provide a shared empirical object for several research communities.

---

93. One Benchmark, Different Contributions

The same benchmark could allow each programme to contribute without abandoning its own methods.

Natural Abstraction researchers improve:

V.

Active Inference researchers improve:

F.

Compositional researchers improve:

C.

Representation researchers improve:

M.

Agent Foundations researchers formalize:

O.

SMFT-style research tests:

L^{+},L^{-},U.

This is what a real research federation would look like.

Not:

one unified terminology.

But:

one shared difficult object.

---

94. A Possible Federation Protocol

The research architecture can be summarized as:

Programme-specific model
↓
shared WORLD interface
↓
common benchmark
↓
cross-framework measurement
↓
translation residual
↓
revision of interfaces. (94.1)

This makes collaboration possible without requiring theoretical conversion.

The earlier source explicitly advocated this style of engagement: begin from a programme's known strength, identify a shared open problem, introduce SMFT only as a candidate interface, and proceed toward test or collaboration. 探討 SMFT 串聯其它名門正派成為五行流轉的可行性

That principle generalizes beyond outreach.

It is a methodology for the research programme itself.

---

95. The Federation Is Not a Pyramid

Another important clarification:

There is no hierarchy:

SMFT

↓
five subordinate schools.

Nor:

Topos
↓
all others.

Nor:

Active Inference
↓
all intelligence.

The architecture is closer to:

a network with cyclic handoffs

plus:

a vertical revision process.

No node is epistemically sovereign.

This is consistent with the broader objective of building systems that remain corrigible under residual evidence.

---

96. The WORLD as a Meeting Object

A useful concept from interdisciplinary work is a shared object that different communities can approach through different languages.

Here the object is:

$\boldsymbol{\mathcal{W}}^{R} = (V,F,C,M,O,L^{+},L^{-},U;P).$ (96.1)

Each programme sees a different projection:

$\pi_i(\boldsymbol{\mathcal{W}}^{R}).$ (96.2)

Thus:

Framework_i studies $\pi_i(\boldsymbol{\mathcal{W}}^{R}).$ (96.3)

The research problem becomes:

> Can the projections be connected strongly enough to reconstruct more of the underlying object?

This is perhaps the cleanest formal picture of the entire article.

---

97. Partial Views Rather Than Competing Totalities

Under this interpretation:

Natural Abstraction sees:

$\pi_NA(\mathcal{W}^R) \approx$ V and its stability.

Active Inference sees:

$\pi_AI(\mathcal{W}^R) \approx$ F plus belief/action relations.

Compositional modeling sees:

$\pi_C(\mathcal{W}^R) \approx$ C and frame transformations.

Representation Geometry sees:

$\pi_M(\mathcal{W}^R) \approx$ M and structural signatures.

Agent Foundations sees:

$\pi_AF(\mathcal{W}^R) \approx$ O and embedded constraints.

SMFT sees:

$\pi_SMFT(\mathcal{W}^R) \approx$ Declaration/Trace/Residual/Revision relations.

None necessarily sees the whole object.

But together they may provide complementary projections.

---

98. The Reconstruction Problem

Suppose the WORLD is latent.

Each research programme observes only a projection:

$y_i = \pi_i(\mathcal{W}) + \varepsilon_i.$ (98.1)

The federation's task is to reconstruct:

$\mathcal{W} = R(y_1,\ldots,y_n).$ (98.2)

This resembles a multi-view inference problem.

No single view need be sufficient.

The reconstruction succeeds if:

$\text{Loss}(\mathcal{W},\mathcal{W}) < \text{Loss}(R(y_i),\mathcal{W})$ (98.3)

for individual views.

This suggests a formal meaning for the value of theoretical pluralism:

> multiple partial frameworks may jointly identify structure that no single projection can recover reliably.

---

99. Cross-Framework Residual Becomes Valuable

Again

If reconstructed predictions fail, the federation receives residual:

r_F = observation – prediction($\mathcal{W}$). (99.1)

That residual may indicate:

a bad local theory,

a bad translation,

a missing WORLD coordinate,

or:

an incorrect federation architecture.

Thus residual does not belong only to the modeled agent.

It belongs to the scientists as well.

This symmetry is philosophically attractive but also methodologically practical.

---

100. From Research Federation to Scientific WORLD Formation

At this point a broader interpretation becomes possible.

Science itself may be understood as constructing effective WORLDS through:

distinction,

dynamics,

formal composition,

measurement,

and embedded observers.

Different research traditions specialize in different parts of this operation.

Scientific progress then occurs partly through:

translation,

residual,

and revision.

The present article does not need to turn this into a universal philosophy of science.

But the analogy helps explain why the federation architecture feels natural:

> the methodology used to study self-revising worlds may itself need to be self-revising.

101. The Core Positioning Result

We can now state the positioning of the five anchor programmes in one compact form:

Natural Abstraction
asks how a bounded WORLD obtains non-arbitrary distinctions.

Active Inference
asks how those distinctions participate in prediction, belief, action, and consequence.

Compositional / Topos approaches
ask how local models and contexts can constitute a coherent formal WORLD.

Representation Geometry / Mechanistic Measurement
asks whether the proposed WORLD structure has measurable realization.

Agent Foundations / Embedded Agency
asks what changes when the observer is itself a bounded part of the WORLD.

Five-Regime SMFT / Boundary Circulation
asks how such a WORLD accumulates historical trace and residual strongly enough to decide when ordinary adaptation must become WORLD revision.

This is the article's positioning map.

---

102. The Core Linkage Result

The stronger result is not the positions themselves.

It is the handoffs:

Natural Abstraction
→ supplies V

Active Inference
→ extends V to (V,F)

Compositional Modeling
→ extends this to (V,F,C)

Representation Measurement
→ extends this to (V,F,C,M)

Agent Foundations
→ extends this to (V,F,C,M,O)

Embedded boundedness
→ regenerates V′.

Thus:

$$V \to F \to C \to M \to O \to V'. \quad (102.1)$$

This is the proposed **World-Bearing Loop**.

---

103. The Core Novelty Result

The novelty of the positioning framework is therefore not:

> five schools correspond to five boxes.

It is:

> **their strongest complementarities may form a productive closure whose output is an Effective WORLD.**

And the deeper extension is:

> **once Trace, Residual, Ledger, and Revision are added, that Effective WORLD becomes a Self-Revising WORLD.**

Symbolically:

$$(V,F,C,M,O) \rightarrow \mathcal{W} \rightarrow (L^{+},L^{-}) \rightarrow U \rightarrow \mathcal{W}'. \quad (103.1)$$

That is the architecture we were seeking.

---

104. From Schools to Worlds

The phrase “From Schools to Worlds” can now be given a precise meaning.

The schools are not dissolved.

Their differences are preserved.

But their outputs are connected.

As the interfaces close, a larger object appears:

not another school,

but:

a WORLD capable of supporting all of their questions simultaneously.

Then, once the WORLD is historically accountable:

not merely a WORLD,

but:

a sequence of WORLDS.

That transition:

$$\mathcal{W}_0 \rightarrow \mathcal{W}_1 \rightarrow \mathcal{W}_2 \rightarrow \dots \quad (104.1)$$

is where the research federation meets the theory of recursive world formation.

105. Toward the Final Sections

The remaining article should now do four things:

1. state clearly what this framework does **not** claim;
2. compare it with simpler alternatives;
3. propose a concrete research programme;
4. conclude with the difference between **theory unification** and **world closure**.

The most important point is already established:

> **The goal is not to make the schools agree. It is to make their interfaces productive enough that their specialized advances can close around a shared object.**

And that object is:

a WORLD.

Appendix A — Positioning the Major Research Programmes

The positioning proposed in the main text should be understood as a map of **dominant research emphasis**, not an exclusive allocation of territory.

Let:

$\mathcal{W} = (V,F,C,M,O).$ (A.1)

where:

V = effective distinctions / abstractions

F = operational dynamics

C = compositional coherence

M = measurable realization

O = embedded observer relation.

A qualitative positioning matrix is:

Research programme	V	F	C	M	O	Dominant interface
Natural Abstraction	High	Medium	Medium	Medium	Medium	environment → effective variables
Active Inference	Medium	High	Medium	Medium	Medium	variables → inference/action
Compositional / Topos approaches	Medium	Medium	High	Low–Medium	Medium	local models → coherent formal world

Research programme	V	F	C	M	O	Dominant interface
Representation Geometry / Mechanistic Measurement	Medium	Medium	Medium	High	Medium	formal structure → measurable realization
Agent Foundations / Embedded Agency	Medium	Medium	Medium	Low–Medium	High	measurable world → bounded internal perspective
SMFT / Boundary Circulation	Cross-cutting	Cross-cutting	Cross-cutting	Cross-cutting	Cross-cutting	WORLD → history → residual → revision

These ratings are intentionally qualitative.

They should eventually be replaced by explicit criteria or empirical measurements.

The important feature is that the matrix is **not diagonal**.

Each programme overlaps several WORLD coordinates.

That is expected.

Appendix B — The Five Productive Handoffs

The proposed World-Bearing Loop contains five particularly important interfaces.

B.1 Natural Abstraction → Active Inference

Output:

effective distinctions V.

Open problem:

how those distinctions become operationally consequential.

Supplement:

inference, prediction, action, and policy.

Structural transition:

$V \to (V,F).$ (B.1)

B.2 Active Inference → Compositional World Modeling

Output:

operative generative dynamics.

Open problem:

how local models, states, contexts, and transformations constitute one coherent formal WORLD.

Supplement:

composition, interfaces, admissibility, frame structure.

Structural transition:

(V,F) → (V,F,C). (B.2)

B.3 Compositional World Modeling → Representation Geometry

Output:

formal WORLD structure.

Open problem:

whether the proposed structure is actually instantiated.

Supplement:

measurement of subspaces, trajectories, basins, transitions, persistent representations, or mechanistic structure.

Structural transition:

(V,F,C) → (V,F,C,M). (B.3)

B.4 Representation Geometry → Agent Foundations

Output:

measurable internal organization.

Open problem:

which structures are available to the agent itself rather than only to an external researcher.

Supplement:

embeddedness, self-reference, bounded reasoning, internal accessibility.

Structural transition:

(V,F,C,M) → (V,F,C,M,O). (B.4)

B.5 Agent Foundations → Natural Abstraction

Output:

bounded embedded observer.

Open problem:

the observer cannot represent its environment in full.

Supplement:

coarse-graining and abstraction.

Structural transition:

$O \rightarrow V'$. (B.5)

Hence:

$V \rightarrow F \rightarrow C \rightarrow M \rightarrow O \rightarrow V'$. (B.6)

This closes the proposed World-Bearing Loop.

Appendix C — The World-Bearing Loop in One Diagrammatic Equation

The full constitutive progression is:

$E \rightarrow V \rightarrow (V,F) \rightarrow (V,F,C) \rightarrow (V,F,C,M) \rightarrow (V,F,C,M,O)=\mathcal{W}$. (C.1)

Embedded boundedness then generates:

$\mathcal{W} \rightarrow O \rightarrow V'$. (C.2)

Therefore:

$\mathcal{W}_n \rightarrow V_{n+1} \rightarrow \ldots \rightarrow \mathcal{W}_{n+1}$. (C.3)

This is why the architecture should be understood as a spiral rather than a closed static ring.

The observer produced within one WORLD becomes part of the conditions from which the next WORLD is abstracted.

Appendix D — One Failure Event, Several Framework Descriptions

Consider the shared event:

E^* = persistent inadequacy of the current effective WORLD. (D.1)

The following descriptions should not be treated as synonyms, but they may refer to partially overlapping structural aspects of E^*.

Framework	Possible description of E^*
Natural Abstraction	current coarse-graining no longer preserves the distinctions needed for prediction or control
Active Inference	persistent mismatch remains under the current generative model despite ordinary updating

Framework	Possible description of E*
Compositional / Topos approaches	the current frame no longer composes the relevant local models coherently
Representation Geometry	trajectories leave a stable representational basin or undergo structural reorganization
Agent Foundations	the embedded agent's current self/world representation no longer supports adequate bounded reasoning
SMFT	structured residual accumulates until the current Declaration can no longer remain stably latched

The research task is not to declare these equivalent.

It is to determine when a partial translation exists.

Let:

$T_{i\to j}(E_i) \approx E_j.$ (D.2)

Then define translation residual:

$\varepsilon_{i\to j} = \mathrm{Loss}[T_{i\to j}(E_i), E_j].$ (D.3)

A useful interface has:

$\varepsilon_{i\to j}$

small enough to preserve the structure relevant to the shared experiment.

Appendix E — What Must Be Preserved Across Framework Translation?

Cross-framework translation should preserve at least one measurable invariant.

Candidate invariants include:

- transition ordering;
- intervention response;
- stability class;
- persistence time;
- information retained;
- recovery behavior;
- identity continuity.

Let I_i denote an invariant expressed in Framework i.

A successful translation should approximately satisfy:

$I_j[T_{i\to j}(E_i)] \approx I_i(E_i).$ (E.1)

Without such preservation, the apparent translation may be only verbal resemblance.

This distinction is central to the proposed federation.

Appendix F — Constitutive Closure and Recursive Closure

The article distinguishes two fundamentally different types of completion.

F.1 Constitutive Closure

An Effective WORLD exists when:

$\mathcal{W} = (V,F,C,M,O)$ (F.1)

is sufficiently coherent for a bounded observer to operate within it.

This may be represented schematically as:

$C_{const}(\mathcal{W}) \geq \Theta_{const}$. (F.2)

Constitutive Closure answers:

> Is there enough structure for a WORLD to operate?

F.2 Recursive Closure

A persistent WORLD additionally requires:

L^+ = admitted historical trace,

L^- = unresolved residual,

U = revision operator.

Thus:

$\mathcal{W}^R = (\mathcal{W},L^+,L^-,U;P)$. (F.3)

Recursive Closure answers:

> Can this WORLD preserve the consequences of its own operation strongly enough to become another WORLD?

Schematically:

$C_{rec}(\mathcal{W}^R) \geq \Theta_{rec}$. (F.4)

The distinction is:

Constitutive Closure → WORLD exists.

Recursive Closure → WORLD can revise itself.

Appendix G — Positioning SMFT More Precisely

SMFT should not be positioned as a sixth research school added to the five-programme ring.

Its proposed role is different.

The five anchor programmes mainly constrain:

$\mathcal{W} = (V,F,C,M,O)$. (G.1)

SMFT primarily focuses on:

$\mathcal{W}_n$
→ Observation
→ Trace / Residual
→ Ledger
→ Latching
→ Revision
→ $\mathcal{W}_{n+1}$. (G.2)

Thus:

research federation → mainly horizontal WORLD constitution;

SMFT → mainly vertical WORLD revision.

This yields:

Horizontal Constitution × Vertical Revision = Recursive World Formation. (G.3)

That is the most defensible positioning of SMFT within the federation.

Appendix H — Relationship to the Five-Regime Runtime

The World-Bearing Loop:

$V \to F \to C \to M \to O \to V'$ (H.1)

should not be identified with the Five-Regime runtime:

$G \to A \to C_r \to S \to R \to G'$. (H.2)

The former describes:

WORLD coordinates and research interfaces.

The latter describes:

dominant runtime transformations.

There is nevertheless an approximate structural resonance:

$V \leftrightarrow G$

$F \leftrightarrow A$

$C \leftrightarrow C_r$

$M \leftrightarrow S$

O ↔ R/G boundary.

This correspondence should be classified as a hypothesis or structural analogy, not as a derivation.

Appendix I — The Research-Federation Falsification Checklist

The positioning framework should be revised if any of the following repeatedly occur.

I.1 The proposed WORLD coordinates are redundant

If V,F,C,M,O cannot be interventionally distinguished, the tuple is over-decomposed.

I.2 The handoffs add no value

If supplying structured output from one programme does not improve the neighboring problem:

$S_{ij} \approx 0, \quad (I.1)$

the proposed interface is weak.

I.3 The cycle does not close

If bounded embeddedness does not systematically regenerate abstraction pressure:

$O \nrightarrow V', \quad (I.2)$

the World-Bearing Loop becomes a chain rather than a cycle.

I.4 One framework is sufficient

If a single framework achieves comparable coverage and prediction at lower total explanatory cost, federation is unnecessary.

I.5 Measurement does not belong in WORLD constitution

If M proves purely epistemic for external scientists rather than constitutive of effective worldhood, $\mathcal{W}$ should be revised.

I.6 Residual adds no predictive value

If L^- does not improve structural-shift detection or recovery, the recursive extension weakens.

The framework must preserve these possibilities.

Appendix J — A Minimal Experimental Programme

A practical research programme could proceed through five increasingly difficult experiments.

J.1 Abstraction–Dynamics Experiment

Test whether stronger natural abstractions improve:

prediction,

transfer,

and control.

Hypothesis:

$$\text{Naturalness}(V) \uparrow \Rightarrow \text{Robustness}(F|V) \uparrow. \quad \text{(J.1)}$$

J.2 Dynamics–Frame Experiment

Create both:

parameter shifts

and:

structural shifts.

Test whether the system can distinguish:

$x \to x'$

from:

$D \to D'$. (J.2)

J.3 Frame–Geometry Experiment

Test whether formal frame changes predict measurable internal reorganization.

Candidate signatures:

subspace shifts,

basin transitions,

hysteresis,

dimensional changes.

J.4 Geometry–Embeddedness Experiment

Test whether externally measurable representation Z is actually available to the agent:

$$\text{External}(Z) \neq \text{Accessible}_O(Z) \quad \text{in general.} \quad \text{(J.3)}$$

J.5 Embeddedness–Abstraction Experiment

Vary:

memory,

compute,

observation bandwidth,

planning depth.

Measure:

$V = V(\text{resource budget}).$ (J.4)

If boundedness systematically changes the abstractions used by the agent, the closing edge gains support.

Appendix K — A Research-Federation Ledger

A practical federation could maintain two records.

Admitted Interface Ledger

L^{+}_Fed = translations and handoffs supported by theory or experiment. (K.1)

Residual Interface Ledger

L^{-}_Fed = failed mappings, disagreements, missing invariants, and unresolved contradictions. (K.2)

The federation itself then becomes revisable:

$Map_{n+1} = U_Fed(Map_n, L^{+}_Fed, L^{-}_Fed).$ (K.3)

This provides an operational safeguard against forced unification.

Appendix L — Final Positioning Summary

The five anchor programmes can be summarized through five questions.

Natural Abstraction

> **What distinctions deserve to exist?**

Active Inference

> **What consequences follow through those distinctions?**

Compositional World Modeling

> **What makes those distinctions and consequences belong to one coherent WORLD?**

Representation Geometry

> **Where is that WORLD actually realized?**

Agent Foundations

> **How can a bounded observer reason from inside that WORLD?**

And the closing question returns to the beginning:

> **Given its bounded perspective, what distinctions can the observer afford to maintain?**

Thus:

Distinction → Consequence → Coherence → Realization → Perspective → Distinction. (L.1)

The cycle closes.

SMFT then asks a second question:

> **What happens when the closed WORLD itself becomes inadequate?**

Its candidate answer is:

WORLD
→ Trace
→ Residual
→ Ledger
→ Revision
→ New WORLD. (L.2)

Final Closing Remark

The deepest proposal of this companion article is therefore not that five contemporary schools secretly belong to one hidden doctrine.

It is almost the opposite.

Their differences may be useful precisely because they prevent one descriptive language from monopolizing the object.

Natural Abstraction can challenge arbitrary variables.

Active Inference can challenge inert representations.

Compositional mathematics can challenge incoherent assemblies.

Representation measurement can challenge purely formal constructions.

Agent Foundations can challenge impossible God's-eye observers.

Residual-driven revision can challenge the WORLD that survived all of them yesterday.

Their complementarities therefore need not culminate in a new master school.

They may culminate in something more useful:

> **a jointly constrained WORLD that no one school is allowed to define alone.**

And when that WORLD fails, the federation does not need to collapse.

It can ask where the failure occurred, preserve what remains valid, retain what cannot yet be explained, and reconstruct the WORLD again.

That is the difference between **unifying theories** and **forming worlds**.

The first tries to make many languages become one.

The second asks whether many partial languages can close around one object strongly enough for that object to become operational, measurable, inhabitable—and revisable.

© 2026 Danny Yeung. All rights reserved. 版权所有 不得转载

Disclaimer

This book is the product of a collaboration between the author and OpenAI's GPT 5.6, Google AI, Gemini 3.X, NoteBookLM, X's Grok, Claude' Sonnet 5 language model. While every effort has been made to ensure accuracy, clarity, and insight, the content is generated with the assistance of artificial intelligence and may contain factual, interpretive, or mathematical errors. Readers are encouraged to approach the ideas with critical thinking and to consult primary scientific literature where appropriate.

This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.